

Team Devdoots: Python Assignment 01 Master Guide

Role: Technical Project Manager (TPM)

Target Audience: All Devdoots Team Members (Balram, Ritesh, Khushi, Harsh Rajbhar, Harsh Gautam)

Objective: Master Python backend logic, data persistence, and professional GitHub PR workflows without merge conflicts.

Phase 0: Prerequisites (Pehle Yeh Padhein aur Seekhein)

Assignment shuru karne se pehle sabhi members ko niche diye gaye concepts clear hone chahiye. Inhe padhe bina code likhna shuru na karein:

- **Variables & Data Types:** Data ko store karne ke liye containers (jaise strings, integers, floats, booleans).
- **Conditions (If-Else):** Program ko decision-making sikhane ke liye rules lagana.
- **Loops & Lists:** Ek sath kai saare items ko store karna aur loop (`for`, `while`) chala kar unhein automate karna.
- **Functions:** Reusable code blocks banana jo value calculate karke `return` karte hain.
- **Dictionaries (JSON):** Key-value pairs ke form mein complex data structure ko manage karna.
- **Error Handling (Try-Except):** Program ko crash hone se bachane ke liye safe-guards lagana.
- **File Handling:** Data ko permanent store karne ke liye files mein write (`"w"`, `"a"`) aur read (`"r"`) karna.

Phase 1: Assignment Problem Statement

Aapko ek **"Smart Inventory & Audit Logger"** Python script banani hai jo niche diye gaye rules ko follow kare:

1. Kam se kam 3 products ki ek list banayein jisme har product ek **Dictionary** ho (`name`, `price`, `stock`).
2. Ek **Function** banayein (`calculate\_total\_inventory\_value()`) jo loop ka use karke total inventory value (`price \* stock`) calculate kare aur return kare.
3. Program mein **Try-Except (Error Handling)** ka use karein taaki agar koi invalid input aaye toh system crash na ho.

4. Calculate ki gayi total value ko ek text file ('inventory_report.txt') mein **File Handling** ke zariye permanent save karein, aur fir wapass read karke terminal par print karein.

Phase 2: Official GitHub Assignment Workflow (The Rules)

Chunki hum sab ek hi repository par kaam kar rahe hain, merge conflicts (code takrana) se bachne ke liye yeh strict rules follow karna anivarya hai [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf]:

Step	Action / Rule Description
Step 1: Pull Latest	Terminal par sabse pehle git pull origin main run karein taaki latest updates mil jayein [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf].
Step 2: Create Branch	Apne naam ki nayi branch banayein [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf]. Format: git checkout -b user/name-assign1 (Example: user/balram-assign1) [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf].
Step 3: Personal File Naming	Folder Assignment_01_Python_Inventory/ ke andar koi bhi common file (jaise main.py ya start.py) nahi banayega [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf]. Har member apne naam se file banayega (jaise balram_inventory.py) [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf].
Step 4: Commit with Tag	Kaam pura hone par proper commit tag use karein [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf]. Format: git

Step	Action / Rule Description
	commit -m "[HW-1] name completed inventory task" [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf].
Step 5: Pull Request (PR)	Code push karke GitHub par Pull Request raise karein [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf]. TPM (Devdoot) code review karega, galtiyen batayega, aur 100% sahi hone par hi main branch mein merge karega [cite: Team Devdoots_ Official GitHub Assignment Workflow.pdf].

Summary of Git Commands (Terminal Flow)

```
git pull origin main
git checkout -b user/yourname-assign1
# Write your code in yourname_inventory.py inside the assignment folder
git add .
git commit -m "[HW-1] yourname finished inventory task"
git origin push user/yourname-assign1
# Go to GitHub and open a Pull Request for TPM review
```